Width-independent Bayesian neural-network learning on continuous input domains

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Abstract

We prove sample-complexity bounds for Bayesian tanh networks with inputs in any compact subset of the Euclidean unit ball, including an entire sphere. A real-valued target is implemented by a teacher of width C, depth L, and weight bound B. For every learner of the same depth and width $N \geq C$, the sample bound is polynomial in d, C, B at fixed depth, independently of N. The result holds as the prior scale tends to zero at each fixed learner width, and at the finite scales specified by every exponent $\gamma \geq 1$, including mean-field scaling. Gaussian weights and a family of smooth, strongly log-concave non-Gaussian weights are covered. Training observes clean real-valued labels and conditions on fitting them within a positive tolerance ρ; testing uses a larger tolerance $\tau > \rho$. The full posterior assigns probability greater than $2/3$ to a prediction within τ on all test inputs outside a set of probability at most ε, with high probability over the sample. No restriction on the input distribution beyond its support is needed, and no global output-margin condition is imposed on the prior. The continuous-domain argument uses a finite input net only inside the proof. Bounded-energy compression handles the strong-prior and $\gamma > 1$ regimes. A bounded dual representation of an arbitrary readout extends the mean-field covariance argument from signs to real outputs. We also obtain exact interpolation in an iterated Gaussian strong-prior limit, and binary classification on supports where a teacher has a uniform positive margin. Exact equality conditioning at fixed prior scale, unbounded teacher weights, and margin-free binary-label classification are not covered.

1 Introduction

A small neural-network implementation can be a useful source of statistical complexity even when the Bayesian learner has many more parameters. This paper establishes such a guarantee on continuous input domains. The sample requirement depends on the width and weight magnitude of a teacher, rather than on the width of the learner. The learner and teacher have the same depth. The dependence on that depth is allowed to be large: throughout the paper, depth is fixed.

There is a basic obstacle to directly importing a Boolean-domain model. A continuous real-valued network whose output is bounded away from zero everywhere on a connected domain has a constant sign. Thus conditioning a prior to have an absolute margin at every point of a connected sphere would exclude every nonconstant binary classifier. We instead start from independent weight coordinates and condition only on the observed training constraints. Our principal result concerns real-valued regression, which permits nonconstant targets on the whole sphere without a margin assumption. For example, $f(x) = \tanh(x_1)$ on S^{d-1} is covered by a width-one, depth-two teacher.

*Authors are listed in alphabetical order. The main learning statements have Lean proofs; see the repository for scope and exclusions.

The labels in this theorem are clean, exact values of the target, but the likelihood accepts a fit within a fixed positive tolerance. Test error means exceeding a strictly larger tolerance. This difference between training and test thresholds provides room for finite approximation. Conditioning directly on exact equality of continuously distributed real outputs at fixed prior scale is a different problem. We nevertheless obtain an exact-interpolation corollary by taking the strong-prior limit first and then sending the training tolerance to zero, in that specified order.

We study two ways to set the prior scale. In the strong-prior limit, the scale tends to zero with learner width held fixed. In the finite-scale family, it is a prescribed power of learner width. The exponent one is the mean-field endpoint; larger exponents give smaller prior scales. Equations (4) and (5) specify the normalization separately for each layer. The sample bound is independent of learner width in both cases, although the strong-prior scale threshold may depend on that width.

The proofs have three parts. First, a neighborhood of a cloned teacher has enough prior mass, with an exponential rate independent of the learner width (the mass itself may decrease with width). Cloning means duplicating teacher neurons and dividing their outgoing weights so that the wider network computes the same function. Second, a finite family approximates all but a sufficiently small amount of prior mass. Third, a union bound shows that an inaccurate member of this family is unlikely to fit the data. For continuous domains, Lipschitz control lets us use a finite input net whose logarithmic cardinality is polynomial in the relevant parameters. The net is a proof device; the data distribution remains arbitrary and continuous inputs are not discretized by the learning rule.

Several probabilistic ingredients extend the finite-input analysis in [1]. We give their statements and the necessary derivations here, and identify where that paper contains a longer proof of a shared compression calculation. The new continuous-domain reductions, posterior arguments, teacher-tube construction, and approximate-readout duality are proved below.

Section 3 gives the two posterior estimates used throughout. Sections 4 and 5 construct the teacher neighborhood and the bounded-energy cover. Section 6 combines them for the strong-prior limit and exponents above one. Section 7 proves the mean-field endpoint using covariance approximation instead of removing small matrix components. Sections 8 and 9 discuss exact interpolation and classification.

2 Model and main results

Fix $d \geq 1$, $L \geq 2$, and a nonempty compact set $X \subseteq \{x \in \mathbb{R}^d : \|x\|_2 \leq 1\}$. Write $\bar{x} = (1, x)$. A network of width N and depth L computes

$$h^0(x) = \bar{x}, \qquad h^\ell(x) = \tanh(W^\ell h^{\ell-1}(x)) \quad (1 \leq \ell < L), \qquad z_w(x) = v^T h^{L-1}(x).$$

All hidden layers have width N, there are no separate internal biases, and the readout is linear. Let $f : X \to \mathbb{R}$ be computed by a teacher of the same depth, width $C \geq 1$, and weights of absolute value at most $B \geq 1$. Width and weight magnitude are separate parameters. The theorem does not claim that width alone controls arbitrary unbounded-weight teachers.

For a raw parameter prior μ, inputs x_i sampled independently from any Borel probability distribution D on X, and exact labels $f(x_i)$, define the posterior by

$$\Pi(dw \mid S_n) = \frac{\prod_{i=1}^n \mathbf{1}\{|z_w(x_i) - f(x_i)| < \rho\}\, \mu(dw)}{\int \prod_{i=1}^n \mathbf{1}\{|z_u(x_i) - f(x_i)| < \rho\}\, \mu(du)}, \qquad \rho > 0. \tag{1}$$

There is no prior conditioning on unobserved inputs. The denominator is positive because a neighborhood of the teacher clone fits uniformly. For $\tau > \rho$, put

$$R_\tau(\Pi) = \mathbb{E}_{w\sim\Pi} D\{|z_w - f| > \tau\}, \qquad q_\tau(x) = \Pi\{|z_w(x) - f(x)| \leq \tau \mid S_n\}.$$

Tonelli and Markov give the posterior-confidence implication

$$D\{q_\tau \le 2/3\} \le 3R_\tau(\Pi). \tag{2}$$

The result controls the whole posterior, including mixtures, rather than only its mean output or a selected parameter minimizer.

Our coordinate law ν has an even variance-one density

$$q(t) = Z^{-1}e^{-V(t)}, \qquad V \in C^2(\mathbb{R}), \qquad 0 < m \le V''(t) \le M < \infty. \tag{3}$$

The law is fixed as width varies. Standard Gaussian coordinates are the case $m = M = 1$. A non-Gaussian example is the variance-standardized density proportional to $\exp\{-t^2/2 - b\log\cosh t\}$ for any fixed $b > 0$. All scalar coordinates below are independent copies of $\xi \sim \nu$. The strong-prior scales are

$$W_{ij}^1 = \sigma\sqrt{N/d}\,\xi_{ij}^1, \qquad W_{ij}^\ell = \sigma\xi_{ij}^\ell \ (2 \le \ell < L), \qquad v_j = \sigma N^{-1/2}\xi_j^L. \tag{4}$$

The finite-scale family sets $\sigma = N^{-\gamma/2}$:

$$W_{ij}^1 = N^{(1-\gamma)/2}\xi_{ij}^1/\sqrt{d}, \quad W_{ij}^\ell = N^{-\gamma/2}\xi_{ij}^\ell, \quad v_j = N^{-(\gamma+1)/2}\xi_j^L. \tag{5}$$

At $\gamma = 1$, the three variances are $1/d$, $1/N$, and $1/N^2$. For compact notation write

$$u = \frac{1}{\min\{1, \tau - \rho\}}, \qquad v_\rho = \log\frac{1}{\min\{1, \rho\}}.$$

Theorem 2.1 (Strong-prior regression)**.** *For every fixed L and coordinate law (3), there is a polynomial p_L such that the following holds. For every teacher as above, every $N \ge C$, every $0 < \rho < \tau$, and every $\varepsilon \in (0,1)$, there is $\sigma_0 > 0$ such that, for all $0 < \sigma < \sigma_0$, all distributions D on X, all $\delta \in (0,1)$, and*

$$n \ge \frac{6}{\varepsilon}\left[p_L(d, C, B, u) + \log\frac{1}{\delta}\right],$$

the posterior (1) satisfies, with probability at least $1 - \delta$ over the sample,

$$R_\tau(\Pi) \le \varepsilon/3, \qquad D\{q_\tau \le 2/3\} \le \varepsilon.$$

The polynomial is independent of learner width. The scale threshold may depend on that finite width, the teacher, ρ, and ε; it does not depend on D, n, δ.

Theorem 2.2 (Finite-scale regression, including mean field)**.** *Fix L, a coordinate law (3), and $\gamma \ge 1$. There is a polynomial $p_{L,\gamma}$ such that, for every teacher as above, every $N \ge C$, every D on X, all $0 < \rho < \tau$, and $\varepsilon, \delta \in (0,1)$, the condition*

$$n \ge \frac{6}{\varepsilon}\left[p_{L,\gamma}\left(d, C, B, u, v_\rho, \log\frac{1}{\varepsilon}\right) + \log\frac{6}{\varepsilon\delta}\right] \tag{6}$$

implies the same risk and posterior-confidence bounds with sample probability at least $1 - \delta$. The polynomial is independent of N. Its degree and constants may depend on L, γ and the coordinate law; no uniformity as $\gamma \downarrow 1$ is claimed for the $\gamma > 1$ proof. The endpoint has a separate proof.

These theorems are finite-width statistical statements. They assert no efficient method for sampling the posterior, no polynomial dependence on depth, and no theorem for $\gamma < 1$. The polynomial degree can itself grow quickly with depth; we do not assert a bound with merely an exponential depth prefactor. A different fixed confidence level below one follows by replacing the factor 3 in (2).

Remark 2.3 (Any bounded coordinate domain). The same conclusions hold for every compact $X \subseteq [-1,1]^d$. All stability and rounding estimates below already use only coordinate bounds. In the input-net estimates replace the unit-ball radius by $\sqrt{d}$, so a mesh r needs at most $(1+2\sqrt{d}/r)^d$ points. Its logarithm still has a polynomial majorant. This includes finite input sets as well as balls and spheres.

3 Two posterior estimates

Both estimates compare the prior mass of inaccurate fitted networks with the mass of a neighborhood that fits every possible sample. The first estimate uses a finite approximation family; the second uses only a lower bound on this neighborhood's mass. Let

$$U_\rho = \{w : \sup_{x \in X} |z_w(x) - f(x)| < \rho\}. \tag{7}$$

This event consists of networks whose outputs stay within the training tolerance of the teacher throughout the domain; we call it a *teacher tube*. It is open in parameter space, since X is compact. Its prior mass lower-bounds the posterior denominator for every sample.

Lemma 3.1 (Learning from a finite approximate cover). *Suppose $\mu(U_\rho) > 0$, a measurable event G obeys $\mu(G^c)/\mu(U_\rho) \le b$, and every output from G is within a uniformly on X of a member of a deterministic finite family $\mathcal{H}$. If $2a < \tau - \rho$ and*

$$n \ge t^{-1}[\log|\mathcal{H}| + \log(1/\delta)],$$

then $R_\tau(\Pi) \le t + b$ with sample probability at least $1-\delta$.

Proof. For each $h \in \mathcal{H}$, set $A_h = \{x : |h(x) - f(x)| > \rho + a\}$. If $D(A_h) > t$, its probability of containing no training input is at most e^{-nt}. A union bound excludes all such surviving h with the asserted probability. Every fitted $w \in G$ has an approximant h whose set A_h contains no training input. Moreover $|z_w - f| > \tau$ implies $x \in A_h$, by $\tau - a > \rho + a$. Thus its test risk is at most t. For every sample, $\Pi(G^c \mid S_n) \le \mu(G^c)/\mu(U_\rho) \le b$. Integrating establishes the conclusion. □

Lemma 3.2 (Learning from a uniform tube). *If $\mu(U_\rho) \ge e^{-A}$, then $n \ge t^{-1}[A + \log(1/(b\delta))]$ implies $R_\tau(\Pi) \le t + b$ with sample probability at least $1-\delta$.*

Proof. A fixed parameter with test risk greater than t fits all stricter training constraints with probability at most e^{-nt}. Tonelli gives expected raw mass of such fitted parameters at most e^{-nt}. Divide by the deterministic denominator lower bound e^{-A} and apply Markov to bound their posterior mass by b with probability $1-\delta$. The remaining parameters each have risk at most t. □

All covers constructed below are finite collections of Borel functions. The relevant exceptional events are measurable: for each such function, the uniform distance from a network output is a Borel function of the parameters, and the constructions alternatively supply explicit measurable norm and covariance-certificate events. No measurable choice of approximant or of teacher-network parameters is needed.

4 Cloning and the mass of a teacher tube

Define the weighted energy

$$\kappa_N(w) = \frac{d}{N}\|W^1\|_F^2 + \sum_{\ell=2}^{L-1}\|W^\ell\|_F^2 + N\|v\|_2^2. \tag{8}$$

Partition each learner layer into C groups with balanced sizes $k_j \in \{\lfloor N/C\rfloor, \lceil N/C\rceil\}$. Copy the teacher neuron in every position of its group, divide an internal teacher edge by its source-group size, and divide each readout edge by its group size. The clone computes f exactly for every real input. Since $\max k_i/\min k_i \le 2$ and $N/\min k_i \le 2C$, its energy is at most

$$E_* := d(d+1)B^2 + 2(L-1)C^2B^2. \tag{9}$$

In particular, the first-layer contribution is at most $d(d+1)B^2$, each internal contribution at most $2C^2B^2$, and the readout contribution at most $2C^2B^2$. These estimates are independent of N.

We use three consequences of (3). A product of ν coordinates conditioned on a centrally symmetric convex set is centered and has

$$\mathbb{E}e^{t\langle u,Z\rangle} \le e^{t^2\|u\|^2/(2m)}. \tag{10}$$

Indeed convex conditioning preserves m-strong log-concavity. The Brascamp–Lieb variance inequality after every linear tilt bounds the Hessian of the log moment generating function by $m^{-1}I$; symmetry sets its derivative at zero to zero. Smooth convex barriers justify conditioning on sets. For the unconditioned product, Caffarelli's contraction theorem and Gaussian isoperimetry give

$$\mu(A_r) \ge \Phi(\Phi^{-1}(\mu(A)) + \sqrt{m}\, r). \tag{11}$$

Here A_r is Euclidean enlargement. Finally any symmetric event E and translation a satisfy

$$\mu(E+a) \ge \mu(E)e^{-M\|a\|^2/2}. \tag{12}$$

To prove the last inequality, translate the conditional law on E. Its relative entropy against μ is $-\log\mu(E) + \mathbb{E}[\sum_i (V(Z_i+a_i) - V(Z_i)) \mid E]$. Taylor's theorem bounds the added expectation by $M\|a\|^2/2$: the linear score term vanishes by oddness of V' and symmetry. A law supported on $E+a$ has relative entropy at least $-\log\mu(E+a)$. References for the two concentration theorems are [3, 4, 5].

Proposition 4.1 (A uniform teacher tube at every width)**.** *For $N \ge C$ and $\gamma \ge 1$,*

$$\mu_{N,\gamma}(U_\rho) \ge e^{-A_L(d,C,B,\rho)N^\gamma}, \tag{13}$$

where A_L has a polynomial majorant in d, C, B, v_ρ at fixed depth and coordinate law.

Proof. Let P project onto the group-constant subspace of the clone, with basis $u_j = k_j^{-1/2}\mathbf{1}_{\text{group } j}$. Put $t = N^{(1-\gamma)/2} \le 1$. In standardized coordinates impose the centered cylinder consisting of $|\xi| \le \eta$ for first-layer and readout entries, and $|\langle Z_{i,*}, u_j\rangle| \le \eta$ for every internal row and group. These scalar constraints concern independent variables, because groups are disjoint.

Every normalized group sum is even, log-concave, and has variance at most $1/m$. Symmetric unimodality and Chebyshev yield, for $0 < \eta \le 1$,

$$\mathbb{P}(|Y| \le \eta) \ge c_m\eta, \qquad c_m = \tfrac{3}{8}\min\{1, \sqrt{m}\}.$$

For example take the interval of radius $2\max\{1, m^{-1/2}\}$, which has mass at least $3/4$, and compare average densities on concentric intervals. Hence the cylinder has mass at least $(c_m\eta)^{N[d+2+(L-2)C]}$.

Conditional on the cylinder, (10) and two $1/4$ sphere nets of size at most 9^N give

$$\mathbb{P}(\|Z\|_{\mathrm{op}} > 2q\sqrt{N} \mid \text{cylinder}) \le 2e^{(2\log 9 - mq^2/2)N}.$$

Set $b_0 = 2\sqrt{(2/m)[2\log 9 + \log(4L)]}$. Intersect the cylinder with $\|Z\|_{\mathrm{op}} \le b_0\sqrt{N}$ for every internal layer. The resulting symmetric event E has at least half the cylinder's mass.

Translate by the standardized clone. The internal perturbation is $tZ/\sqrt{N} = tZP/\sqrt{N} + tZ(I - P)/\sqrt{N}$. Its second summand annihilates the teacher activations, and its first has norm at most $\eta\sqrt{C}$. If $e_\ell = N^{-1/2}\|h^\ell - h_*^\ell\|_2$, boundedness and contraction of tanh give, uniformly on X,

$$e_1 \le \eta(d+1), \qquad e_\ell \le (\sqrt{2}CB + b_0)e_{\ell-1} + \eta\sqrt{C}.$$

Put $A_0 = 2 + 3CB + b_0$. Then $e_{L-1} \le \eta(d+L+1)A_0^{L-2}$. The clone has $\sqrt{N}\|v_*\| \le \sqrt{2}CB$; the perturbation readout has ℓ^1 norm at most η. Its output error is therefore at most ηP_0, where

$$P_0 = 1 + 2CB(d+L+1)A_0^{L-2}, \qquad \eta = \min\{1/2, \rho/(2P_0)\}.$$

The translated event lies in U_ρ. Its standardized translation has squared norm $N^\gamma\kappa_N(w_*) \le E_*N^\gamma$. Equation (12) and $N \le N^\gamma$ prove the claim with, for instance,

$$A_L = 1 + \tfrac{M}{2}E_* + [d + 2 + (L-2)C]\log\frac{1}{c_m\eta} + \log 2. \tag{14}$$

All stability estimates used only bounded input coordinates, so they hold simultaneously on the continuous domain. □

5 Uniform covers for bounded energy

We now construct a finite family that approximates every bounded-energy network uniformly on the input domain, regardless of learner width. The following finite-input calculation is the deterministic compression ingredient shared with [1]. Only the logarithm of the number of input points enters it.

Lemma 5.1 (Finite-input neuron compression)**.** *Let $T \subset [-1,1]^d$ be finite of cardinality s, $K \ge 1$, and $0 < a \le 1$. Every network with $\kappa_N(w) \le K$ has a width-J approximant with energy at most $4K$ and error at most $a/4$ on T, where J has a polynomial majorant in $d, K, 1/a, \log(2s)$ at fixed L. One may take*

$$\lambda = \frac{a2^{-L}K^{-(L-1)/2}}{4(L-1)}, \quad D_T = \max\{1, \lceil\log_2 s\rceil\}, \quad J = \max\left\{\lceil 4K\rceil, \left\lceil 256\lambda^{-4}[D_T + \log(L/\lambda)]\right\rceil\right\}.$$

All weights of the approximant have magnitude at most J.

Proof. Here are the estimates underlying the neuron-sampling construction; the full reverse-order telescoping calculation is also given in the bounded-energy compression appendix of [1]. In each internal or readout row of Euclidean norm r_i, delete entries exceeding $2r_i/(\lambda\sqrt{N})$. Their total absolute contribution against a bounded hidden vector is at most $(\lambda/2)\sqrt{N}r_i$. At every hidden layer independently sample J uniform neuron indices with replacement. Retain the sampled first-layer rows; multiply each retained edge out of a sampled neuron by N/J.

For each retained row and input, Hoeffding's inequality bounds the probability that the estimated sum differs from the truncated sum by more than $(\lambda/2)\sqrt{N}r_i$ by $2e^{-\lambda^4 J/32}$. Union bounding over inputs, rows, and layers costs at most $2sLJe^{-\lambda^4 J/32} \le 1/100$. The expected weighted energy of the sampled network is at most K; independence of adjacent layers' index lists suffices for the internal matrix calculation. Thus its energy is at most $4K$ with probability at least $3/4$. The sum of the squared norms of retained rows, with the N normalization, is at most $8JK$ with probability at least $7/8$, again by Markov. These three events have a nonempty intersection.

Fix a realization in that intersection and expose sampled layers in reverse order. The downstream matrices at each step already equal their final sampled matrices, whose norms are at most $2\sqrt{K}$, and the sampled readout has norm at most $2\sqrt{K/J}$. Propagating the row errors shows that each truncation and each sampling stage changes the output by at most $a/[8(L-1)]$. The $2(L-1)$ stages total at most $a/4$. This is precisely why the choice of λ contains $2^{-L}K^{-(L-1)/2}$. The energy bound gives entrywise bounds $\sqrt{4KJ/d}$, $2\sqrt{K}$, and $2\sqrt{K/J}$ at the three layer types, each at most J since $J \ge 4K$. □

Proposition 5.2 (A continuous-domain energy cover)**.** *For fixed L, $K \ge 1$, and $0 < a \le 1$, there is a deterministic finite family $\mathcal{H}_{K,a}$ of Borel functions on X, common to all learner widths, such that*

$$\inf_{h \in \mathcal{H}_{K,a}} \sup_X |z_w - h| \le a \quad \textit{whenever } \kappa_N(w) \le K,$$

and $\log|\mathcal{H}_{K,a}| \le H_L(d, K, 1/a)$ for a polynomially bounded function at fixed depth.

Proof. Multiplying the layer norm bounds from (8) gives

$$\mathrm{Lip}(z_w) \le K^{L/2}/\sqrt{d}. \tag{15}$$

Choose an input net $T \subset X$ of radius $r = \min\{1, a\sqrt{d}/(8K^{L/2})\}$ and size $s \le (1 + 2/r)^d$. Fix a nearest-net-point map $t : X \to T$, resolving ties by a fixed enumeration. It is Borel. Apply Lemma 5.1 on T.

For any width-J network with weights bounded by J, rounding every weight to a grid of mesh

$$\Delta \le \frac{a}{4(d + L + 1)(2J^2)^L}$$

changes its output by at most $a/4$ whenever the input coordinates are bounded by one. To check this, a first-layer coordinate error is at most $(d+1)\Delta$; each next coordinate error is at most $2J^2$ times the preceding error plus $J\Delta$, including the readout. Induction gives the displayed bound. There are $P = J(d+1) + (L-2)J^2 + J$ coordinates, so at most $(1 + 2(J+1)/\Delta)^P$ rounded networks.

Use their functions $x \mapsto z_{\mathrm{rounded}}(t(x))$ as cover centers. Compression, rounding, and (15) give total error at most $a/4 + a/4 + a/8 < a$. The centers need not themselves be continuous or neural networks; they are used only in the statistical union bound. Since $\log s = O_L(d \log(2 + K + 1/a))$, the logarithm of their number has a fixed-depth polynomial majorant and is independent of N. □

6 The strong-prior and super-mean-field proofs

At fixed width and small prior scale, networks with excessive energy have negligible posterior mass. At exponents above one, the analogous finite-scale argument first removes matrix components with small singular values. Both routes lead to the same bounded-energy cover.

Proof of Theorem 2.1. Fix N and the exact clone w_*. The tube U_ρ is an open neighborhood of w_*. Evenness and (3) give two-sided quadratic density envelopes $ce^{-Mt^2/2} \le q(t) \le C_0 e^{-mt^2/2}$. In the finite-dimensional parameter space, elementary Gaussian-integral bounds therefore yield

$$\liminf_{\sigma\downarrow 0} \sigma^2 \log \mu_\sigma(U_\rho) \ge -\tfrac{M}{2} E_*, \qquad \limsup_{\sigma\downarrow 0} \sigma^2 \log \mu_\sigma\{\kappa_N > K\} \le -\tfrac{m}{2} K.$$

For the lower bound integrate a sufficiently small open neighborhood of w_* whose energy is at most $E_* + \zeta$, then let $\zeta \downarrow 0$. For the upper bound integrate the quadratic envelope outside the ellipsoid; fixed dimension contributes only subexponential factors. Take $K = (M/m)E_* + 2$. The exceptional-to-tube mass ratio tends to zero. Choose σ_0 making it at most $\varepsilon/6$ for all smaller scales. Apply Proposition 5.2 with $a = \min\{1/2, (\tau - \rho)/4\}$ and then Lemma 3.1 with $t = b = \varepsilon/6$. Equation (2) finishes the proof. □

The same proof uses only independent coordinate densities with common two-sided envelopes $c_j e^{-c_+ t^2} \le q_j(t) \le C_j e^{-c_- t^2}$, where $0 < c_- \le c_+ < \infty$ do not depend on width. Symmetry and log-concavity are unnecessary in this strong-prior extension. Replace K by $(c_+/c_-)E_* + 2$; prefactors affect only the fixed-width scale threshold.

Lemma 6.1 (Spectral truncation at $\gamma > 1$)**.** *Fix $K \ge 1$ and $0 < a < 1$, and put $b = a/[2L(\sqrt{K} + 1)^{L-2}]$. If*

$$N^{\gamma-1} \ge 8(d+1)/m, \qquad N^{\gamma-1} \ge 256/(mb^2), \tag{16}$$

then, except with probability $Le^{-mKN^\gamma/8}$, the network output is uniformly within $a/2$ on X of a network of energy at most LK.

Proof. Equation (10) implies $\mathbb{E}e^{m\xi^2/4} \le \sqrt{2}$ by integrating against a one-dimensional Gaussian. Hence, for D_0 independent coordinates,

$$\mathbb{P}\left(\sum_{j=1}^{D_0} \xi_j^2 > u\right) \le 2^{D_0/2} e^{-mu/4} \le e^{-mu/8} \quad \text{if } mu \ge 8D_0. \tag{17}$$

This controls the first-layer weighted norm and the readout weighted norm by K outside two events of probability $e^{-mKN^\gamma/8}$.

For an internal matrix $A = \sigma Z$, the net bound gives $\mathbb{P}(\|A\|_{\mathrm{op}} \le 8\sigma\sqrt{N/m}) \ge 1/2$. If $b \ge 16\sigma\sqrt{N/m}$, the operator-norm ball of radius $b/2$ has mass at least $1/2$. Its Frobenius distance from A is $D(A)^2 = \sum_j (s_j(A) - b/2)_+^2$. Equation (11) implies $\mathbb{P}(D(A) > r) \le e^{-mr^2/(2\sigma^2)}$. The singular part $A_{>b}$ has $\|A_{>b}\|_F \le 2D(A)$, whence

$$\mathbb{P}(\|A_{>b}\|_F^2 > K) \le e^{-mK/(8\sigma^2)}.$$

At $\sigma = N^{-\gamma/2}$ the second condition in (16) permits this estimate.

Discard all internal singular values at most b. The retained network has energy at most LK. If e_ℓ denotes its Euclidean hidden error, $e_1 = 0$ and $e_\ell \le \sqrt{K} e_{\ell-1} + b\sqrt{N}$. The readout norm is at most $\sqrt{K/N}$, so the uniform output error is at most $b\sum_{j=1}^{L-2} K^{j/2} \le a/2$. The $L = 2$ sum is empty. A union bound proves the assertion. □

Proof of Theorem 2.2 when $\gamma > 1$. Let $A = A_L(d, C, B, \rho)$, $a = \min\{1/2, (\tau - \rho)/4\}$, and

$$K = \max\{1, (8/m)[A + \log(6L/\varepsilon)]\}.$$

Let T be the ceiling of the maximum of C and the two thresholds in (16). For $N \geq T$, Lemma 6.1 and a cover of the retained networks to accuracy $a/2$ give a common finite output cover of accuracy a. By (13), its exceptional-to-tube mass ratio is at most $Le^{-N^\gamma(mK/8-A)} \leq \varepsilon/6$. Apply Lemma 3.1. For $C \leq N < T$, use Lemma 3.2 with tube cost at most AT^γ. At fixed $L, \gamma > 1$, the cover entropy and AT^γ have polynomial majorants in the arguments of (6). This covers every $N \geq C$. □

7 The mean-field endpoint

At $\gamma = 1$ the typical internal matrices have nonvanishing operator norm. Discarding their bulk does not produce a good approximation. Instead, finite covariance tests describe most prior outputs, and a bounded dual representation turns covariance approximation into output approximation. The argument below retains the actual first-layer law. Auxiliary Gaussian rows are used to construct finite descriptions; the learner's posterior is not replaced by a Gaussian process.

7.1 A bounded dual for an arbitrary readout

On a finite input net, the hidden activations form a matrix and the readout is a linear combination of its columns. The next lemma approximates that output using the activation covariance instead. Its coefficient bound controls the output error when the covariance is subsequently approximated.

Lemma 7.1 (Approximate-readout duality)**.** *Let $H \in \mathbb{R}^{s\times N}$, $Q = HH^T/N$, $z = Hv$, and $N\|v\|_2^2 \leq K$. For every $t > 0$ there exists $\beta \in \mathbb{R}^s$ such that*

$$\|Q\beta - z\|_\infty \leq t, \qquad \|\beta\|_1 \leq \frac{K}{4t}.$$

Proof. Minimize $N\|u\|_2^2/2$ subject to $\|Hu - z\|_\infty \leq t$. The point v is strictly feasible. Convex optimality gives a signed difference β of the lower and upper multipliers with $Nu = H^T\beta$. Complementarity gives

$$t\|\beta\|_1 = \beta^T(z - Hu) = Nv^Tu - N\|u\|^2.$$

The two multipliers for the same coordinate cannot both be positive, because $t > 0$. Put $r = \sqrt{N}\|u\|$. The right side is at most $\sqrt{K}r - r^2 \leq K/4$. Also $Hu = Q\beta$, proving both claims. Neither a sign margin nor invertibility of Q is required. □

7.2 Finite covariance certificates

We test an empirical activation covariance against quadratic exponential moments of its reference feature law. Passing all tests with a bounded cost supplies a nearby covariance described by a bounded exponential tilt. The probability estimate and the approximation conclusion are the two parts of the next lemma. For a law μ on $[-1, 1]^s$, a symmetric matrix Γ, and an empirical covariance $\widehat{Q}$, write

$$\mathcal{J}_G(\widehat{Q}; \mu) = \max_{\|\Gamma\|_{1,\mathrm{ent}} \leq G} \left\{ \langle \Gamma, \widehat{Q} \rangle - \log \mathbb{E}_\mu e^{h^T\Gamma h} \right\}, \qquad \|\Gamma\|_{1,\mathrm{ent}} = \sum_{i,j} |\Gamma_{ij}|.$$

The norm dual to this entrywise ℓ^1 norm is $\|A\|_{\max} = \max_{i,j} |A_{ij}|$.

Lemma 7.2 (Distribution-free certificate)**.** *For an empirical second moment of N independent samples from μ,*

$$\mathbb{P}\{\mathcal{J}_G(\widehat{Q}; \mu) > K\} \leq \exp\{q \log(2s^2 + 1) - N(K - 1)\}, \qquad q = \lceil G^2/2 \rceil. \tag{18}$$

If $\mathcal{J}_G \le K$, some quadratic exponential tilt of μ, with parameter norm at most G, has covariance within K/G of $\widehat{Q}$ in maximum norm.

Proof. Flatten the s^2 bounded observables $F_{ij} = h_i h_j$ into a vector $F \in [-1,1]^D$, and write $\Lambda(\lambda) = \log \mathbb{E} e^{\lambda \cdot F}$. Every ℓ^1-ball vector λ is the mean of a random vector in $\{0, \pm G e_1, \dots, \pm G e_D\}$. Average q independent such vectors to get a finite set $\mathcal{V}$ of size at most $(2D+1)^q$. Hoeffding's exponential-moment bound and Jensen imply $\mathbb{E}\Lambda(V) \le \Lambda(\lambda) + G^2/(2q)$. Thus for every moment vector a,

$$\max_{\|\lambda\|_1 \le G} \{\lambda \cdot a - \Lambda(\lambda)\} \le 1 + \max_{v \in \mathcal{V}} \{v \cdot a - \Lambda(v)\}.$$

For empirical a, independence gives $\mathbb{E} e^{N[v \cdot a - \Lambda(v)]} = 1$. Markov and the finite union bound prove (18).

For the approximation assertion, let λ_* maximize the dual and let μ_* be the corresponding exponential tilt, with moments a_*. The first-order condition for maximization over the ℓ^1 ball gives $\lambda_* \cdot (a - a_*) = G\|a - a_*\|_\infty$. Therefore

$$\mathcal{J}_G = \mathrm{KL}(\mu_* \| \mu) + G\|a - a_*\|_\infty.$$

Relative entropy is nonnegative. Symmetrizing the parameter matrix does not change its quadratic form or increase its norm. □

For $Q \succeq 0$, let $\mathsf{F}_\Gamma(Q)$ be the covariance of $h = \tanh Z$, $Z \sim \mathcal{N}(0, Q)$, tilted by $e^{h^T \Gamma h}$. Two bounds control this Gaussian auxiliary family.

Lemma 7.3 (Relative comparison and covariance stability)**.** *Let $\|a_j\|_\infty \le N^{-1/2}$, let $\xi_j \sim \nu$ and $g_j \sim \mathcal{N}(0,1)$ independently, and put $Z_\nu = \sum_j a_j \xi_j$, $Z_G = \sum_j a_j g_j$. For $\|\Gamma\|_{1,\mathrm{ent}} \le G$ and $N \ge 4G^2$,*

$$\left| \log \mathbb{E} e^{\tanh(Z_\nu)^T \Gamma \tanh(Z_\nu)} - \log \mathbb{E} e^{\tanh(Z_G)^T \Gamma \tanh(Z_G)} \right| \le \frac{c_\nu P_3(G)}{\sqrt{N}}, \tag{19}$$

where $P_3(G) = 24G + 36G^2 + 8G^3$ and $c_\nu = (e/6)(\mathbb{E}|\xi|^3 e^{|\xi|} + \mathbb{E}|g|^3 e^{|g|}) < \infty$. Also, for all positive semidefinite Q, Q',

$$\|\mathsf{F}_\Gamma(Q) - \mathsf{F}_\Gamma(Q')\|_{\max} \le 8(1+G)^2 \|Q - Q'\|_{\max}. \tag{20}$$

Neither bound depends on the number s of field coordinates.

Proof. Put $F(z) = \tanh(z)^T \Gamma \tanh(z)$. In a direction $\|a\|_\infty \le N^{-1/2}$ its first three directional derivatives are bounded by $2G/\sqrt{N}$, $6G/N$, and $24G/N^{3/2}$. Consequently $|D_a^3 e^F| \le P_3(G) N^{-3/2} e^F$. At a one-coordinate replacement, fix the residual z and define $f_z(t) = e^{F(z+at)}$. Since $2G/\sqrt{N} \le 1$,

$$e^{-|t|} f_z(0) \le f_z(t) \le e^{|t|} f_z(0).$$

Taylor expansion through order two and matched mean and variance imply that the two averaged values differ by at most $P_3(G) N^{-3/2} (\mathbb{E}|\xi|^3 e^{|\xi|} + \mathbb{E}|g|^3 e^{|g|}) f_z(0)/6$. Each average is at least $e^{-1} f_z(0)$ by Jensen and variance one. These bounds persist after averaging over the residual. The inequality $|\log u - \log v| \le |u - v| / \min(u, v)$ and N replacements give (19). This relative calculation avoids an exponential loss in the tilt budget.

For (20), interpolate covariance linearly and differentiate the Gaussian tilted expectation of $H(z) = \tanh(z_i)\tanh(z_j)$, using the quotient rule. If M_k denotes the supremum of the sum of absolute derivatives of order k, then $M_1(F) \le 2G$, $M_2(F) \le 6G$, $M_1(H) \le 2$, and $M_2(H) \le 6$. Gaussian differentiation bounds the derivative by

$$\tfrac{1}{2}\|Q - Q'\|_{\max}[6 + 8G + 2(6G + 4G^2)] \le 8(1+G)^2 \|Q - Q'\|_{\max}.$$

All terms are expectations or covariances under the normalized tilted law, so no reciprocal partition-function estimate is used. Integrate the derivative. Adding ηI and taking $\eta \downarrow 0$ justifies singular covariance matrices. Expanded calculations for both estimates appear in [1]. □

7.3 Most outputs have a common finite cover

Proposition 7.4 (Mean-field output cover)**.** *There are fixed-depth polynomially bounded functions $T_L(d,K,1/a)$ and $H_L^{\mathrm{mf}}(d,K,1/a)$ with the following property. For $0<a<1$ and K above a fixed-law constant at least 4, there is a finite deterministic family $\mathcal{C}_{K,a}$, independent of learner width, such that for every $N \ge T_L(d,K,1/a)$,*

$$\log|\mathcal{C}_{K,a}| \le H_L^{\mathrm{mf}}(d,K,1/a), \qquad \mu_{N,1}\left\{\inf_{h\in\mathcal{C}_{K,a}} \sup_X |z_w - h| > a\right\} \le 2Le^{-cNK},$$

where $c = \min\{1/4, m/16\}$.

Proof. We give the construction, including its width and net dependencies.

Lipschitz event. Require

$$\|W^1\|_F^2 \le NK, \qquad \|W^\ell\|_{\mathrm{op}} \le \sqrt{K}\ (2\le\ell<L), \qquad N\|v\|^2 \le K.$$

On this event $\mathrm{Lip}(z_w) \le K^{L/2}$. The square moment bound (17) gives first-layer and readout failure at most $e^{-mNK/8}$ for sufficiently large fixed-law K. For the first layer the exponent before simplification is $-mdNK/4 + N(d+1)\log 2/2$. The sphere-net estimate gives internal failure at most $2e^{2N\log 9 - mNK/8} \le e^{-mNK/16}$ after enlarging the same constant baseline. Thus total failure is at most $Le^{-mNK/16}$.

Choose a net $S \subset X$ of radius $a/(4K^{L/2})$ and cardinality

$$s \le (1 + 8K^{L/2}/a)^d. \tag{21}$$

Fix a Borel nearest-net-point map $t : X \to S$. The actual output changes by at most $a/4$ between x and $t(x)$ on the Lipschitz event.

Proxy covariance tower. Write $r = L-1$, $\bar{K} = K+1$, and

$$G_r = 8rK\bar{K}/a^2, \qquad G_\ell = 8G_{\ell+1}(1+G_{\ell+1})^2 \quad (\ell<r), \qquad b_\ell = 8(1+G_\ell)^2.$$

Let $\ell_0 = \log 3 + 2d\log(1+8K^{L/2}/a)$, so that $\log(2s^2+1) \le \ell_0$ uniformly over compact X in the unit ball. For $X \subseteq [-1,1]^d$ as in Remark 2.3, replace $8K^{L/2}/a$ by $8\sqrt{d}\,K^{L/2}/a$ here and in (21). Take

$$T_L = \left\lceil \max\left\{1, \frac{2\lceil G_1^2/2\rceil \ell_0}{K-1}, 4G_1^2, c_\nu^2 P_3(G_1)^2\right\}\right\rceil. \tag{22}$$

These budgets and this threshold have polynomial majorants in $d, K, 1/a$ at fixed L, using (21).

Let $Q_{\ell,N}$ be the empirical activation covariance indexed by S. At the first layer, apply Lemma 7.2 to its actual non-Gaussian field law μ_0. At later layers apply it conditionally to the true row law. A union bound bounds failure by $re^{-NK/4}$. An internal preactivation is $N^{-1/2}\sum_j \xi_j h_j$, so (19) compares its log partition functions to the Gaussian law of covariance $Q_{\ell-1,N}$ with error at most one under (22). The Gaussian certificate is therefore at most $\bar{K}$. The lift in Lemma 7.2 produces matrices Γ_ℓ with entrywise norm at most G_ℓ. Define Q_1^* by tilting the actual first-layer law with Γ_1, and recursively set $Q_\ell^* = \mathsf{F}_{\Gamma_\ell}(Q_{\ell-1}^*)$ for $\ell \ge 2$. Equation (20) gives

$$\|Q_{r,N} - Q_r^*\|_{\max} \le \sum_{\ell=1}^{r} \frac{\bar{K}}{G_\ell} \prod_{j=\ell+1}^{r} b_j = \frac{r\bar{K}}{G_r} = \frac{a^2}{8K}.$$

Readout and finite realization. On the readout norm event, Lemma 7.1 with tolerance $a/4$ gives $\|\beta\|_1 \le K/a$ and $\|Q_{r,N}\beta - z_w|_S\|_\infty \le a/4$. Replacing the covariance by Q_r^* adds error at most $a/8$.

Here is a finite network realizing the proxy to another $a/8$. Put $b = 8(1+G_1)^2$ and

$$\eta = \frac{a^2}{8Krb^{r-1}}, \qquad J = \left\lceil 2\eta^{-2}\log(8rs^2)\right\rceil.$$

Sample J first-layer rows from the actual incoming law tilted by $e^{h^T\Gamma_1 h}$. At each later layer, sample independent Gaussian rows divided by $\sqrt{J}$, tilted by $e^{\tanh(Z)^T\Gamma_\ell \tanh(Z)}$ conditionally on the previous sampled activations. Their reference Gaussian field has covariance equal to the previous empirical covariance. Hoeffding and a union bound over rs^2 covariance entries give simultaneous empirical error at most η with probability at least $3/4$. Covariance stability then gives final covariance error at most $rb^{r-1}\eta$. Set each readout coefficient to $(\beta\cdot h_i)/J$. The output error against $Q_r^*\beta$ is at most $(K/a)rb^{r-1}\eta = a/8$.

All sampled weights can simultaneously be bounded polynomially. Each row's tilted density relative to its reference is at most e^{2G_1}. The reference coordinates obey $\mathbb{E}e^{a_0\xi^2} \le \sqrt{2}$ for $a_0 = \min\{m,1\}/4$. If $D_J = J(d+1)+(L-2)J^2$, choose

$$B_0 = \max\left\{1, \frac{K}{aJ}, \sqrt{\frac{2G_1 + \log(4\sqrt{2}\max\{D_J,1\})}{a_0}}\right\}.$$

A union bound gives probability at least $3/4$ that all incoming weights are bounded by B_0. Internal weights are additionally divided by $\sqrt{J}$, so the bound remains valid. The readout bound is deterministic. Intersecting with the covariance event gives a realization with width and weights polynomially bounded in $d, K, 1/a$ at fixed depth.

Counting and extension off the net. For that common width and weight bound, put

$$P = J(d+1)+(L-2)J^2+J, \qquad \Lambda = (d+LJ)[1+J(B_0+1)]^{L-1}.$$

Round parameters to mesh $a/(8\Lambda)$. A layer-by-layer coordinate-error induction bounds the output change by $a/8$ on all inputs with coordinates at most one. There are at most $\exp\{P\log(3+16B_0\Lambda/a)\}$ rounded networks, with logarithm polynomially bounded in the required parameters.

Include the functions $h(x) = z_{\text{rounded}}(t(x))$ in the cover. Its error to the actual output is at most

$$\underbrace{a/4}_{\text{readout dual}} + \underbrace{a/8}_{\text{proxy covariance}} + \underbrace{a/8}_{\text{finite realization}} + \underbrace{a/8}_{\text{rounding}} + \underbrace{a/4}_{\text{input net}} = 7a/8 < a.$$

Only the actual output needs a Lipschitz bound; no circular choice of net based on the approximant occurs. The norm and certificate failures total at most $2Le^{-cNK}$. The grid of possible centers depends on the common bounds and the fixed net, and hence is independent of N and of the draw. □

Proof of Theorem 2.2 when $\gamma = 1$. Let $A = A_L(d,C,B,\rho)$ and $a = \min\{1/2, (\tau-\rho)/4\}$. Choose K above the baseline in Proposition 7.4 and with $cK \ge A + \log(12L/\varepsilon)$. At $N \ge T_L(d,K,1/a)$, the exceptional-to-tube mass ratio is at most $2Le^{-N(cK-A)} \le \varepsilon/6$. Lemma 3.1 gives risk at most $\varepsilon/3$ at sample size $(6/\varepsilon)[H_L^{\text{mf}}(d,K,1/a)+\log(1/\delta)]$. For $C \le N < T_L$, Lemma 3.2 uses the tube cost AT_L. Both bounds have the polynomial majorant in (6). Equation (2) completes the argument at every width $N \ge C$. □

8 Exact interpolation in an iterated strong-prior limit

Positive training tolerance is necessary for the ordinary conditional posterior (1) to be defined by a positive-probability event. There is nonetheless a precise strong-prior exact-interpolation conclusion. It follows from one sample event that controls all low-energy interpolants, including all mixtures of them.

Proposition 8.1 (Every low-energy interpolant generalizes)**.** *Let E_* be the clone energy bound (9), and fix $\tau > 0$. For fixed L, let H be the logarithmic cardinality of the all-width energy cover at energy E_* and radius $a = \min\{1, \tau/8\}$. It is polynomially bounded in $d, C, B, 1/\min\{1, \tau\}$. If*

$$n \geq \frac{3}{\varepsilon}[H + \log(1/\delta)],$$

then, with probability at least $1 - \delta$, simultaneously for every $N \geq C$ and every parameter satisfying

$$\kappa_N(w) \leq E_*, \qquad \max_i |z_w(x_i) - f(x_i)| \leq \tau/2,$$

one has $D\{|z_w - f| > \tau\} \leq \varepsilon/3$. In particular the same bound holds for every minimum-energy exact interpolant and for every probability mixture of such interpolants. All these mixtures satisfy (2).

Proof. For a cover center h set $A_h = \{|h - f| > 3\tau/4\}$. A covered parameter fitting within $\tau/2$ has $|h(x_i) - f(x_i)| \leq \tau/2 + a \leq 5\tau/8$, so all training inputs avoid A_h. Its test errors lie in A_h, since $\tau - a \geq 7\tau/8 > 3\tau/4$. A union bound excludes every surviving center with $D(A_h) > \varepsilon/3$ at the displayed sample size. The cover is common to all widths, so the assertion is simultaneous without a union bound over widths. An exact teacher clone is feasible with energy at most E_*, so every energy-minimizing exact interpolant belongs to the covered sublevel. The risk bound persists under mixtures. □

Corollary 8.2 (An exact Bayesian strong-prior limit)**.** *Use Gaussian coordinates in (4). At every fixed $N \geq C$, every iterated weak limit of the posterior obtained by first sending $\sigma \downarrow 0$ at fixed $\rho > 0$, and then sending $\rho \downarrow 0$, is supported on minimum-energy exact interpolants. Such subsequential limits exist. With the sample bound of Proposition 8.1, all of them obey the stated regression and posterior-confidence guarantees.*

Proof. Fix a clean sample and put $e_S(w) = \max_i |z_w(x_i) - f(x_i)|$, $O_\rho = \{e_S < \rho\}$, $b_\rho = \inf_{O_\rho} \kappa_N$, and $b_0 = \min_{\{e_S = 0\}} \kappa_N$. The exact feasible set is nonempty and closed. Since κ_N is continuous and coercive, its minimum exists, with $b_0 \leq E_*$.

The Gaussian density is proportional to $e^{-\kappa_N/(2\sigma^2)}$. For every fixed $\rho, \eta > 0$, an open ball inside O_ρ has energy below $b_\rho + \eta/2$. Comparing its Gaussian mass to the tail $\{\kappa_N > b_\rho + \eta\}$ proves

$$\Pi_{\sigma,\rho}\{\kappa_N > b_\rho + \eta\} \longrightarrow 0.$$

This gives tightness, since energy sublevels are compact. Each posterior has $e_S \leq \rho$ and $\kappa_N \geq b_\rho$ almost surely, so every inner weak limit is supported on $\{e_S \leq \rho,\ \kappa_N = b_\rho\}$. This does not require identifying the closure of O_ρ with the whole non-strict feasible set.

As $\rho \downarrow 0$, b_ρ increases to b_0. Indeed choose almost minimizers at tolerances tending to zero. They lie in a common compact energy sublevel; a convergent subsequence has exact fit and energy at most $\lim b_\rho$, giving the reverse inequality to $b_\rho \leq b_0$. The inner limiting measures are supported on the common compact set $\{\kappa_N \leq b_0\}$, and every outer weak limit is supported on $\{e_S = 0, \kappa_N = b_0\}$. Proposition 8.1 then applies directly to their support on its single high-probability sample event. □

The limit order in Corollary 8.2 is part of the statement. It does not identify a fixed-scale regular conditional posterior given exact outputs, reverse the two limits, or prove a result along arbitrary joint paths $(\sigma, \rho) \to (0, 0)$. No rank or regular-value assumption on the sample-evaluation map was used. Under two-sided quadratic tail envelopes with exponents $0 < c_- \le c_+$, the analogous safe energy bound is $(c_+/c_-)E_*$, giving the same polynomial conclusion, without identifying minimum-quadratic-energy interpolants.

9 Binary classification and the role of a margin

The continuous-regression theorem does not assume that the teacher output avoids zero. Classification requires a separate denominator argument.

Corollary 9.1 (Classification on a support with a teacher margin)**.** *Let $g : X \to \{-1, 1\}$ and suppose a teacher of width C, depth L, and weight bound B satisfies $g(x)z_*(x) \ge r > 0$ on X. Condition the raw prior only on the training margins $g(x_i)z_w(x_i) > 1/2$. For both the strong-prior regime (at each fixed $N \ge C$, for all σ below a threshold that may depend on N, the teacher, r, and ε, as in Theorem 2.1) and every fixed $\gamma \ge 1$, there are fixed-depth sample bounds polynomial in $d, C, B, 1/r$ and the accuracy parameters, independent of learner width for every $N \ge C$, such that with sample probability at least $1 - \delta$,*

$$D\{x : \Pi(g(x)z_w(x) > 0 \mid S_n) \le 2/3\} \le \varepsilon.$$

The teacher margin may hold on any full-measure support; no assertion about unobserved points outside that support is required.

Proof. Multiply the teacher readout by $3/(2r)$, obtaining signed output at least $3/2$. Its width is still C, and its weight bound is at most $\max\{B, 3B/(2r)\}$. A uniform output tube of radius $1/2$ fits all training margins. The preceding target-tube estimates therefore apply. For the finite-cover step use radius $a < 1/4$. If h approximates a fitted network, every training point satisfies $g(x_i)h(x_i) > 1/2 - a$. Every sign error of that network lies in $\{x : g(x)h(x) \le 1/2 - a\}$, since $a < 1/2 - a$. A union bound over these sets replaces Lemma 3.1; the small-width mass-only argument is unchanged. The risk-to-confidence bound is again Markov. □

For $d \ge 2$, the sphere S^{d-1} is connected. A continuous network with $|z_*| \ge r > 0$ everywhere on it has a constant sign, by the intermediate value theorem. Thus a nonconstant full-sphere binary target does not satisfy the corollary's uniform-margin premise. The corollary is useful, for example, for distributions whose support excludes a band around a decision boundary. The disconnected sphere S^0 is an exception to the connectedness observation.

For a nonconstant example on the entire connected sphere, take the regression target $f(x) = \tanh(x_1)$, represented at depth two and width one with weights bounded by one. Every $N \ge 1$ is covered, for every distribution on the sphere, and any fixed $0 < \rho < \tau$. This example needs no exclusion of an equatorial band. At larger fixed depth, iterated compositions of tanh give corresponding width-one examples.

Real score labels also permit sign prediction without a uniform margin. For any nonzero target value with $|f(x)| > \tau$, the event $|z_w(x) - f(x)| \le \tau$ implies the correct sign. Hence

$$D\{\Pi(\operatorname{sign} z_w = \operatorname{sign} f \mid S_n) \le 2/3\} \le \varepsilon + D\{|f| \le \tau\} \tag{23}$$

on the regression theorem's success event. Any convention for the sign of zero is harmless, since zeros lie in the second set. This is a corollary for training on real-valued scores, not for training on binary labels alone.

For $f(x) = \tanh(x_1)$ and uniform spherical inputs in dimension $d \geq 2$,

$$D\{|f| \leq \tau\} \leq 3\sqrt{d}\,\tau \qquad (0 < \tau \leq 1/2). \tag{24}$$

To check it, let $s = \operatorname{artanh}\tau$. Then $s \leq 4\tau/3$ and $s \leq \operatorname{artanh}(1/2) < 0.55$. For $d \geq 3$, the first-coordinate density is $c_d(1-t^2)^{(d-3)/2}$ with $c_d = \Gamma(d/2)/(\sqrt{\pi}\Gamma((d-1)/2)) \leq \sqrt{d}$; the ratio bound follows by applying Jensen to the square root of a Gamma variable. For $d = 2$, the density $1/(\pi\sqrt{1-t^2})$ is at most one on $[-s, s]$. Integrating yields at most $2s\sqrt{d} \leq 3\sqrt{d}\tau$. Taking regression accuracy $\varepsilon/2$, $\tau = \varepsilon/(6\sqrt{d})$, and $\rho = \tau/2$ gives correct-sign posterior confidence on all but an ε fraction of the entire sphere, with a fixed-depth sample bound polynomial in $d, 1/\varepsilon$, at every learner width $N \geq 1$. The observed labels here remain the real scores $\tanh(x_1)$.

10 What this result does and does not establish

The argument establishes a continuous-input version of width-independent learning with a positive training tolerance. Its sample bound is polynomial in teacher width and weight bound at fixed depth, for every learner width above the teacher width. The distribution may have atoms, a density, or lower-dimensional support; no covering-number assumption on the distribution itself is made. The unit-ball assumption is used to bound the logarithm of an input-net size and the dependence of outputs on parameters. A bounded domain of a different radius can be rescaled, changing the teacher's first-layer weight bound.

The positive tolerance in the finite-scale theorem is substantive. It does not prove exact real-valued interpolation under a regular conditional posterior or a zero-tolerance limit at fixed prior scale. Section 8 supplies a distinct exact-interpolation result with a specified order of limits. In the strong-prior tolerance theorem, the sample polynomial can be independent of the absolute training tolerance, but the scale threshold may deteriorate as that tolerance shrinks. At finite scale the displayed bound tracks this cost through $\log^+(1/\rho)$. The inverse gap $1/(\tau - \rho)$ also enters the polynomial, so the theorem does not make a uniform assertion as the test and training thresholds coincide.

The binary corollary assumes a uniform teacher margin on the data support. Dropping that assumption would require a different lower bound on posterior normalization, or quantitative control of how much input mass lies near the teacher's decision boundary. The present proofs do not establish a width-only, margin-free classification theorem. They also do not establish the bounds for ReLU, noisy labels, $\gamma < 1$, or efficient posterior computation. No robustness restriction on the network prior is imposed, and no polynomial dependence on depth is claimed.

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